

Step 1: Create Database

```
CREATE DATABASE bank_management;  
USE bank_management;
```

Step 2: Create Tables

Customers

```
CREATE TABLE Customers (  
    CustomerID INT PRIMARY KEY,  
    Name VARCHAR(100),  
    DOB DATE,  
    Balance DECIMAL(10,2),  
    IsVIP BOOLEAN DEFAULT FALSE,  
    LastModified DATETIME  
);
```

Accounts

```
CREATE TABLE Accounts (  
    AccountID INT PRIMARY KEY,  
    CustomerID INT,  
    AccountType VARCHAR(20),  
    Balance DECIMAL(10,2),  
    LastModified DATETIME,  
    FOREIGN KEY (CustomerID)  
    REFERENCES Customers(CustomerID)  
);
```

Transactions

```
CREATE TABLE Transactions (  
    TransactionID INT PRIMARY KEY,  
    AccountID INT,  
    TransactionDate DATE,  
    Amount DECIMAL(10,2),  
    TransactionType VARCHAR(20),  
    FOREIGN KEY (AccountID)  
    REFERENCES Accounts(AccountID)  
);
```

Loans

```
CREATE TABLE Loans (  
    LoanID INT PRIMARY KEY,
```

```
CustomerID INT,  
LoanAmount DECIMAL(10,2),  
InterestRate DECIMAL(5,2),  
StartDate DATE,  
EndDate DATE,  
FOREIGN KEY (CustomerID)  
REFERENCES Customers(CustomerID)  
);
```

Employees

```
CREATE TABLE Employees (  
EmployeeID INT PRIMARY KEY,  
Name VARCHAR(100),  
Position VARCHAR(50),  
Salary DECIMAL(10,2),  
Department VARCHAR(50),  
HireDate DATE  
);
```

AuditLog

```
CREATE TABLE AuditLog (  
AuditID INT AUTO_INCREMENT PRIMARY KEY,  
TransactionID INT,  
ActionDate DATETIME,  
ActionTaken VARCHAR(100)  
);
```

Step 3: Insert Sample Data

Customers

```
INSERT INTO Customers VALUES  
(1,'John Doe','1955-05-15',15000,FALSE,NOW()),  
(2,'Jane Smith','1990-07-20',12000,FALSE,NOW()),  
(3,'Robert Wilson','1948-03-10',8000,FALSE,NOW()),  
(4,'David Miller','1985-08-25',25000,FALSE,NOW());
```

Accounts

```
INSERT INTO Accounts VALUES  
(101,1,'Savings',5000,NOW()),  
(102,2,'Checking',12000,NOW()),
```

```
(103,3,'Savings',7000,NOW()),  
(104,4,'Savings',25000,NOW());
```

Transactions

```
INSERT INTO Transactions VALUES  
(1,101,CURDATE(),1000,'Deposit'),  
(2,102,CURDATE(),500,'Withdrawal'),  
(3,103,CURDATE(),700,'Deposit'),  
(4,104,CURDATE(),300,'Withdrawal');
```

Loans

```
INSERT INTO Loans VALUES  
(1,1,100000,10,CURDATE(),DATE_ADD(CURDATE(),INTERVAL 20 DAY)),  
(2,2,200000,12,CURDATE(),DATE_ADD(CURDATE(),INTERVAL 100 DAY)),  
(3,3,150000,11,CURDATE(),DATE_ADD(CURDATE(),INTERVAL 15 DAY)),  
(4,4,300000,9,CURDATE(),DATE_ADD(CURDATE(),INTERVAL 200 DAY));
```

Employees

```
INSERT INTO Employees VALUES  
(1,'Alice Johnson','Manager',70000,'HR','2015-06-15'),  
(2,'Bob Brown','Developer',60000,'IT','2017-03-20'),  
(3,'Charlie Davis','Analyst',50000,'Finance','2019-04-10'),  
(4,'Emma Wilson','Developer',65000,'IT','2020-05-12');
```

Exercise 1: Control Structures

Scenario 1: Customers above 60 get 1% loan interest reduction

DELIMITER //

```
CREATE PROCEDURE ApplySeniorDiscount()  
BEGIN  
    UPDATE Loans l  
    JOIN Customers c  
    ON l.CustomerID=c.CustomerID  
    SET l.InterestRate=l.InterestRate-1  
    WHERE TIMESTAMPDIFF(YEAR,c.DOB,CURDATE())>60;  
END //
```

DELIMITER ;

Run:

```
CALL ApplySeniorDiscount();
```

```
SELECT * FROM Loans;
```

Scenario 2: Promote VIP customers

```
DELIMITER //
```

```
CREATE PROCEDURE PromoteVIP()
```

```
BEGIN
```

```
    UPDATE Customers
```

```
    SET IsVIP=TRUE
```

```
    WHERE Balance>10000;
```

```
END //
```

```
DELIMITER ;
```

Run:

```
CALL PromoteVIP();
```

```
SELECT * FROM Customers;
```

Scenario 3: Loan reminders

```
SELECT CustomerID,
```

```
    LoanID,
```

```
    EndDate
```

```
FROM Loans
```

```
WHERE EndDate
```

```
BETWEEN CURDATE()
```

```
AND DATE_ADD(CURDATE(),INTERVAL 30 DAY);
```

Exercise 2: Error Handling

SafeTransferFunds

```
DELIMITER //
```

```
CREATE PROCEDURE SafeTransferFunds(
```

```
    IN fromAcc INT,
```

```
    IN toAcc INT,
```

```
    IN amount DECIMAL(10,2)
```

```
)
```

```
BEGIN
```

```
    DECLARE bal DECIMAL(10,2);
```

```

SELECT Balance
INTO bal
FROM Accounts
WHERE AccountID=fromAcc;

IF bal<amount THEN
    SIGNAL SQLSTATE '45000'
    SET MESSAGE_TEXT='Insufficient Funds';
ELSE
    UPDATE Accounts
    SET Balance=Balance-amount
    WHERE AccountID=fromAcc;

    UPDATE Accounts
    SET Balance=Balance+amount
    WHERE AccountID=toAcc;
END IF;
END //

```

DELIMITER ;

Run:

```
CALL SafeTransferFunds(101,102,1000);
```

UpdateSalary

DELIMITER //

```

CREATE PROCEDURE UpdateSalary(
    IN empid INT,
    IN percentIncrease INT
)
BEGIN
    UPDATE Employees
    SET Salary=
    Salary+(Salary*percentIncrease/100)
    WHERE EmployeeID=empid;

    IF ROW_COUNT()=0 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT='Employee Not Found';
    END IF;
END //

```

```
END IF;  
END //
```

```
DELIMITER ;
```

Run:

```
CALL UpdateSalary(1,10);
```

Exercise 3: Stored Procedures

Monthly Interest

```
DELIMITER //
```

```
CREATE PROCEDURE ProcessMonthlyInterest()  
BEGIN  
    UPDATE Accounts  
    SET Balance=Balance*1.01  
    WHERE AccountType='Savings';  
END //
```

```
DELIMITER ;
```

Run:

```
CALL ProcessMonthlyInterest();
```

Exercise 4: Functions

Calculate Age

```
DELIMITER //
```

```
CREATE FUNCTION CalculateAge(dob DATE)  
RETURNS INT  
DETERMINISTIC  
BEGIN  
    RETURN TIMESTAMPDIFF(YEAR,dob,CURDATE());  
END //
```

```
DELIMITER ;
```

Run:

```
SELECT CalculateAge('1990-01-01');
```

Check Balance

DELIMITER //

```
CREATE FUNCTION HasSufficientBalance(  
    accid INT,  
    amt DECIMAL(10,2)  
)  
RETURNS BOOLEAN  
DETERMINISTIC  
BEGIN  
    DECLARE bal DECIMAL(10,2);  
  
    SELECT Balance  
    INTO bal  
    FROM Accounts  
    WHERE AccountID=accid;  
  
    RETURN bal>=amt;  
END //
```

DELIMITER ;

Run:

```
SELECT HasSufficientBalance(101,500);
```

Exercise 5: Triggers

Update Last Modified

DELIMITER //

```
CREATE TRIGGER UpdateCustomerLastModified  
BEFORE UPDATE  
ON Customers  
FOR EACH ROW  
BEGIN  
    SET NEW.LastModified=NOW();  
END //
```

DELIMITER ;

Audit Trigger

DELIMITER //

```
CREATE TRIGGER LogTransaction
AFTER INSERT
ON Transactions
FOR EACH ROW
BEGIN
    INSERT INTO AuditLog
    (TransactionID,ActionDate,ActionTaken)
    VALUES
    (NEW.TransactionID,
    NOW(),
    'Transaction Inserted');
END //
```

DELIMITER ;

Transaction Rules Trigger

DELIMITER //

```
CREATE TRIGGER CheckTransactionRules
BEFORE INSERT
ON Transactions
FOR EACH ROW
BEGIN
    DECLARE bal DECIMAL(10,2);

    SELECT Balance
    INTO bal
    FROM Accounts
    WHERE AccountID=NEW.AccountID;

    IF NEW.TransactionType='Withdrawal'
    AND NEW.Amount>bal THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT='Insufficient Balance';
    END IF;

    IF NEW.TransactionType='Deposit'
    AND NEW.Amount<=0 THEN
        SIGNAL SQLSTATE '45000'
```



```

SET MESSAGE_TEXT='Invalid Deposit';
END IF;
END //

DELIMITER ;

```

The screenshot displays the MySQL Workbench interface with two windows. The top window, titled 'sakila-schema', shows a SQL editor with the following queries:

```

1 • CREATE DATABASE bank_management;
2 • USE bank_management;

```

The bottom window, titled 'MySQL Workbench', shows a SQL editor with the following queries:

```

2 CustomerID INT PRIMARY KEY,
3 Name VARCHAR(100),
4 DOB DATE,
5 Balance DECIMAL(10,2),
6 IsVIP BOOLEAN DEFAULT FALSE,
7 LastModified DATETIME
8 );
9 • CREATE TABLE Accounts (
10 AccountID INT PRIMARY KEY,
11 CustomerID INT,
12 AccountType VARCHAR(20),
13 Balance DECIMAL(10,2),
14 LastModified DATETIME,
15 FOREIGN KEY (CustomerID)
16 REFERENCES Customers(CustomerID)
17 );
18 • CREATE TABLE Transactions (
19 TransactionID INT PRIMARY KEY,
20 AccountID INT,
21 TransactionDate DATE,
22 Amount DECIMAL(10,2),
23 TransactionType VARCHAR(20),
24 FOREIGN KEY (AccountID)
25 REFERENCES Accounts(AccountID)
26 );

```

The 'Output' pane at the bottom shows the execution results for the queries in the bottom window:

#	Time	Action	Message	Duration / Fetch
5	15:20:25	CREATE TABLE Accounts (AccountID INT PRIMARY KEY, CustomerID INT, AccountType VARCHAR...	0 row(s) affected	0.062 sec
6	15:20:26	CREATE TABLE Transactions (TransactionID INT PRIMARY KEY, AccountID INT, TransactionDate D...	0 row(s) affected	0.063 sec

MySQL Workbench - Local instance MySQL33 - W...

File Edit View Query Database Server Tools Scripting Help

Navigator

Filter objects

Schemas

- cv2_db
- cv2_db
- insure_db
- mydatabase
- nvd
- product
- Tables
- Views
- Stored Procedures
- Administration
- Schemas

Information

No object selected

Query 1 sakila-schema sakila-data SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* SQL File 11* SQL File 12* SQL File 13*

1 CREATE TABLE Loans (
2 Execute the selected portion of the script or everything, if there is no selection
3 CustomerID INT,
4 LoanAmount DECIMAL(10,2),
5 InterestRate DECIMAL(5,2),
6 StartDate DATE,
7 EndDate DATE,
8 FOREIGN KEY (CustomerID)
9 REFERENCES Customers(CustomerID)
10);
11
12 CREATE TABLE Employees (
13 EmployeeID INT PRIMARY KEY,
14 Name VARCHAR(100),
15 Position VARCHAR(50),
16 Salary DECIMAL(10,2),
17 Department VARCHAR(50),
18 HireDate DATE
19);
20
21 CREATE TABLE AuditLog (
22 AuditID INT AUTO_INCREMENT PRIMARY KEY,
23 TransactionID INT,
24 ActionDate DATETIME,
25 ActionTaken VARCHAR(100)
26);

Output

Action Output

#	Time	Action	Message	Duration / Fetch
8	15:21:00	CREATE TABLE Employees (EmployeeID INT PRIMARY KEY, Name VARCHAR(100), Position VARC...	0 row(s) affected	0.032 sec
9	15:21:00	CREATE TABLE AuditLog (AuditID INT AUTO INCREMENT PRIMARY KEY, TransactionID INT, Acti...	0 row(s) affected	0.031 sec

Object Info Session

35°

Search

ENG IN 15:21:02 30-06-2026

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

MySQL Workbench - Local instance MySQL33 - W...

File Edit View Query Database Server Tools Scripting Help

Navigator

Filter objects

Schemas

- cv2_db
- cv2_db
- insure_db
- mydatabase
- nvd
- product
- Tables
- Views
- Stored Procedures
- Administration
- Schemas

Information

No object selected

Query 1 sakila-schema sakila-data SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* SQL File 11* SQL File 12* SQL File 13*

1 INSERT INTO Customers VALUES
2 (Execute the selected portion of the script or everything, if there is no selection
3 (2,'Jane Smith','1990-07-20',12000,FALSE,NOW()),
4 (3,'Robert Wilson','1948-03-10',8000,FALSE,NOW()),
5 (4,'David Miller','1985-08-25',25000,FALSE,NOW());
6
7 INSERT INTO Accounts VALUES
8 (101,1,'Savings',5000,NOW()),
9 (102,2,'Checking',12000,NOW()),
10 (103,3,'Savings',7000,NOW()),
11 (104,4,'Savings',25000,NOW());
12
13 INSERT INTO Transactions VALUES
14 (1,101,CURDATE(),1000,'Deposit'),
15 (2,102,CURDATE(),500,'Withdrawal'),
16 (3,103,CURDATE(),700,'Deposit'),
17 (4,104,CURDATE(),300,'Withdrawal');
18
19 INSERT INTO Loans VALUES
20 (1,1,100000,10,CURDATE(),DATE_ADD(CURDATE(),INTERVAL 20 DAY)),
21 (2,2,200000,12,CURDATE(),DATE_ADD(CURDATE(),INTERVAL 100 DAY)),
22 (3,3,150000,11,CURDATE(),DATE_ADD(CURDATE(),INTERVAL 15 DAY)),
23 (4,4,300000,9,CURDATE(),DATE_ADD(CURDATE(),INTERVAL 200 DAY));
24
25 INSERT INTO Employees VALUES
26 (1,'Alice Johnson','Manager',70000,'HR','2015-06-15'),
27 (2,'Bob Brown','Developer',60000,'IT','2017-03-20'),
28 (3,'Charlie Davis','Analyst',50000,'Finance','2019-04-10'),
29 (4,'Emma Wilson','Developer',65000,'IT','2020-05-12');

Output

Action Output

#	Time	Action	Message	Duration / Fetch
13	15:22:14	INSERT INTO Loans VALUES (1,1,100000,10,CURDATE(),DATE_ADD(CURDATE(),INTERVAL 20 DAY)) (2...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.000 sec
14	15:22:14	INSERT INTO Employees VALUES (1,'Alice Johnson','Manager',70000,'HR','2015-06-15') (2,'Bob Brown','Deve...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.015 sec

Object Info Session

35°

Search

ENG IN 15:22:16 30-06-2026

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

```
Query 1 sakila-schema sakila-data SQL File 5 SQL File 6 SQL File 7 SQL File 8 SQL File 9 SQL File 10 SQL File 11 SQL File 12 SQL File 13
Limit to 1000 rows
1 DELIMITER //
2
3 CREATE PROCEDURE ApplySeniorDiscount()
4 BEGIN
5     UPDATE Loans l
6     JOIN Customers c
7     ON l.CustomerID=c.CustomerID
8     SET l.InterestRate=l.InterestRate-1
9     WHERE TIMESTAMPDIFF(YEAR,c.DOB,CURDATE())>60;
10 END //
11
12 DELIMITER ;
```

MySQL Workbench

Local instance MySQL83 - W...

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- cvr_db
- cv2_db
- insural_db
- mydatabase
- nvd
- product
- Tables
- Views
- Stored Procedures

Administration Schemas

Information

No object selected

Query 1 sakila-schema sakila-data SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* SQL File 11* SQL File 12* SQL File 13*

1

2 SELECT * FROM Loans;

Result Grid

	LoanID	CustomerID	LoanAmount	InterestRate	StartDate	EndDate
1	1	1	100000.00	10.00	2026-06-30	2026-07-30
2	2	2	200000.00	12.00	2026-06-30	2026-10-08
3	3	3	150000.00	11.00	2026-06-30	2026-07-15
4	4	4	300000.00	9.00	2026-06-30	2027-01-15
5	5	5	100000.00	10.00	2026-06-30	2026-07-30

Loans 1 x

Output

Action Output

20 15:23:50 CALL ApplySeniorDiscount()

21 15:24:05 SELECT * FROM Loans LIMIT 0, 1000

Message

Error Code: 1175. You are using safe update mode and you tried to update a table without a WHERE that uses...

4 row(s) returned

Duration / Fetch

0.000 sec

0.000 sec / 0.000 sec

Object Info Session

35°

Q Search

ENG IN

15:24:09 30-06-2026

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQL Workbench

Local instance MySQL33 - W...

File Edit View Query Database Server Tools Scripting Help

Navigator

Schemas

- Filter objects
- cve_db
- cve2_db
- insure_db
- mydatabase
- nvd
- product
- Tables
- Views
- Stored Procedures

Administration Schemas

Information

No object selected

Query 1 sakila-schema sakila-data SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* SQL File 11* SQL File 12* SQL File 13*

1 DELIMITER //

2

3 CREATE PROCEDURE PromoteVIP()

4 BEGIN

5 UPDATE Customers

6 SET IsVIP=TRUE

7 WHERE Balance>10000;

8 END //

9

10 DELIMITER ;

Output

Action Output

Time Action Message Duration / Fetch

21 15:24:05 SELECT * FROM Loans LIMIT 0, 1000 4 row(s) returned 0.000 sec / 0.000 sec

22 15:24:25 CREATE PROCEDURE PromoteVIP() BEGIN UPDATE Customers SET IsVIP=TRUE WHERE Balance... 0 row(s) affected 0.015 sec

Object Info Session

35°

Search

ENG IN 15:24:28 30-06-2026

MySQL Workbench

Local instance MySQL33 - W...

File Edit View Query Database Server Tools Scripting Help

Navigator

Schemas

- Filter objects
- cve_db
- cve2_db
- insure_db
- mydatabase
- nvd
- product
- Tables
- Views
- Stored Procedures

Administration Schemas

Information

No object selected

Query 1 sakila-schema sakila-data SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* SQL File 11* SQL File 12* SQL File 13*

1 CALL PromoteVIP();

2

3 SELECT * FROM Customers;

Result Grid

CustomerID	Name	DOB	Balance	IsVIP	LastModified
1	John Doe	1955-05-15	15000.00	1	2026-06-30 15:22:14
2	Jane Smith	1990-07-20	12000.00	1	2026-06-30 15:22:14
3	Robert Wilson	1948-03-10	8000.00	0	2026-06-30 15:22:14
4	David Miller	1985-08-25	25000.00	1	2026-06-30 15:22:14

Customers 2 x

Output

Action Output

Time Action Message Duration / Fetch

25 15:25:30 CALL PromoteVIP() 3 row(s) affected 0.016 sec

26 15:25:30 SELECT * FROM Customers LIMIT 0, 1000 4 row(s) returned 0.000 sec / 0.000 sec

Object Info Session

35°

Search

ENG IN 15:25:34 30-06-2026

MySQL Workbench - Local instance MySQL33 - W...

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHMAS

Filter objects

- cve_db
- cv2_db
- insure_db
- mydatabase
- nvd
- product
- Tables
- Views
- Stored Procedures

Administration Schemas

Information

No object selected

Query 1 sakila-schema sakila-data SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* SQL File 11* SQL File 12* SQL File 13*

1 SELECT CustomerID,
2 Execute the selected portion of the script or everything, if there is no selection
3
4 EndDate
5 FROM Loans
6 WHERE EndDate
7 BETWEEN CURDATE()
8 AND DATE_ADD(CURDATE(),INTERVAL 30 DAY);

Result Grid

CustomerID	LoanID	EndDate
1	1	2026-07-20
3	3	2026-07-15

Loans 3 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
26	15:25:30	SELECT * FROM Customers LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
27	15:26:28	SELECT CustomerID, LoanID, EndDate FROM Loans WHERE EndDate BETWEEN CURDATE() AN...	2 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

35°

Search

ENG IN 15:26:30 30-06-2026

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQL Workbench - Local instance MySQL33 - W...

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHMAS

Filter objects

- cve_db
- cv2_db
- insure_db
- mydatabase
- nvd
- product
- Tables
- Views
- Stored Procedures

Administration Schemas

Information

No object selected

Query 1 sakila-schema sakila-data SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* SQL File 11* SQL File 12* SQL File 13*

1 SELECT CustomerID,
2 LoanID,
3 EndDate
4 FROM Loans
5 WHERE EndDate
6 BETWEEN CURDATE()
7 AND DATE_ADD(CURDATE(),INTERVAL 30 DAY)DELIMITER //
8
9 CREATE PROCEDURE SafeTransferFunds(
10 IN fromAcc INT,
11 IN toAcc INT,
12 IN amount DECIMAL(10,2)
13)
14 BEGIN
15 DECLARE bal DECIMAL(10,2);
16
17 SELECT Balance
18 INTO bal
19 FROM Accounts
20 WHERE AccountID=fromAcc;
21
22 IF bal<amount THEN
23 SIGNAL SQLSTATE '45000'
24 SET MESSAGE_TEXT='Insufficient Funds';
25 ELSE

Output

Action Output

#	Time	Action	Message	Duration / Fetch
27	15:26:28	SELECT CustomerID, LoanID, EndDate FROM Loans WHERE EndDate BETWEEN CURDATE() AN...	2 row(s) returned	0.000 sec / 0.000 sec
28	15:26:47	CREATE PROCEDURE SafeTransferFunds(IN fromAcc INT, IN toAcc INT, IN amount DECIMAL(10,2)...	0 row(s) affected	0.015 sec

Object Info Session

35°

Search

ENG IN 15:26:51 30-06-2026

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQL Workbench

Local instance MySQL93 - W...

File Edit View Query Database Server Tools Scripting Help

Navigator

Filter objects

Schemas

- cv2_db
- cv2_db
- insure_db
- mydatabase
- nvd
- product
- Tables
- Views
- Stored Procedures

Administration Schemas

Information

No object selected

Query 1 sakila-schema sakila-data SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* SQL File 11* SQL File 12* SQL File 13*

1 CALL SafeTransferFunds(101,102,1000);

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Output

Action Output

#	Time	Action	Message	Duration / Fetch
29	15:27:05	CALL SafeTransferFunds(101,102,1000)	1 row(s) affected	0.032 sec
30	15:27:09	CALL SafeTransferFunds(101,102,1000)	1 row(s) affected	0.015 sec

Object Info Session

35°

Search

ENG IN

15:27:14 30-06-2026

MySQL Workbench

Local instance MySQL93 - W...

File Edit View Query Database Server Tools Scripting Help

Navigator

Filter objects

Schemas

- cv2_db
- cv2_db
- insure_db
- mydatabase
- nvd
- product
- Tables
- Views
- Stored Procedures

Administration Schemas

Information

No object selected

Query 1 sakila-schema sakila-data SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* SQL File 11* SQL File 12* SQL File 13*

```
1 DELIMITER //
2 Execute the selected portion of the script or everything, if there is no selection
3 CREATE PROCEDURE UpdateSalary(
4     IN empid INT,
5     IN percentIncrease INT
6 )
7 BEGIN
8     UPDATE Employees
9     SET Salary=
10     Salary*(Salary*percentIncrease/100)
11     WHERE EmployeeID=empid;
12
13 IF ROW_COUNT()=0 THEN
14     SIGNAL SQLSTATE '45000'
15     SET MESSAGE_TEXT='Employee Not Found';
16 END IF;
17 END //
18
19 DELIMITER ;
```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Output

Action Output

#	Time	Action	Message	Duration / Fetch
30	15:27:09	CALL SafeTransferFunds(101,102,1000)	1 row(s) affected	0.015 sec
31	15:27:31	CREATE PROCEDURE UpdateSalary(IN empid INT, IN percentIncrease INT) BEGIN UPDATE Empl...	0 row(s) affected	0.031 sec

Object Info Session

35°

Search

ENG IN

15:27:34 30-06-2026

MySQL Workbench

Local instance MySQL33 - W...

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- cv2_db
- cv2_db
- insure_db
- mydatabase
- nvd
- product
- Tables
- Views
- Stored Procedures

Administration Schemas

Information

No object selected

Query 1 sakila-schema sakila-data SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* SQL File 11* SQL File 12* SQL File 13*

Limit to 1000 rows

1 CALL UpdateSalary(1,10);

Execute the selected portion of the script or everything, if there is no selection

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Output

Action Output

#	Time	Action	Message	Duration / Fetch
31	15:27:31	CREATE PROCEDURE UpdateSalary(IN empid INT, IN percentincrease INT) BEGIN UPDATE Empl...	0 row(s) affected	0.031 sec
32	15:27:48	CALL UpdateSalary(1,10)	1 row(s) affected	0.016 sec

Object Info Session

35°

Search

ENG IN 15:27:50 30-06-2026

MySQL Workbench

Local instance MySQL33 - W...

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- cv2_db
- cv2_db
- insure_db
- mydatabase
- nvd
- product
- Tables
- Views
- Stored Procedures

Administration Schemas

Information

No object selected

Query 1 sakila-schema sakila-data SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* SQL File 11* SQL File 12* SQL File 13*

Limit to 1000 rows

```
1 DELIMITER //
2 Execute the selected portion of the script or everything, if there is no selection
3 CREATE PROCEDURE ProcessMonthlyInterest()
4 BEGIN
5     UPDATE Accounts
6     SET Balance=Balance*1.01
7     WHERE AccountType='Savings';
8 END //
9
10 DELIMITER ;
```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Output

Action Output

#	Time	Action	Message	Duration / Fetch
32	15:27:48	CALL UpdateSalary(1,10)	1 row(s) affected	0.016 sec
33	15:28:06	CREATE PROCEDURE ProcessMonthlyInterest() BEGIN UPDATE Accounts SET Balance=Balance*1.0...	0 row(s) affected	0.016 sec

Object Info Session

35°

Search

ENG IN 15:28:07 30-06-2026

MySQL Workbench

Local instance MySQL33 - W...

File Edit View Query Database Server Tools Scripting Help

Navigator

Schemas

- Filter objects
- cve_db
- cv2_db
- insure_db
- mydatabase
- nvd
- product
- Tables
- Views
- Stored Procedures

Administration Schemas

Information

No object selected

Query 1 sakila-schema sakila-data SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* SQL File 11* SQL File 12* SQL File 13*

Limit to 1000 rows

```
1 DELIMITER //
2
3 CREATE FUNCTION CalculateAge(dob DATE)
4 RETURNS INT
5 DETERMINISTIC
6 BEGIN
7     RETURN TIMESTAMPDIFF(YEAR,dob,CURDATE());
8 END //
9
10 DELIMITER ;
11 SELECT CalculateAge('1990-01-01');
```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
34	15:28:23	CREATE FUNCTION CalculateAge(dob DATE) RETURNS INT DETERMINISTIC BEGIN RETURN TIMES...	0 row(s) affected	0.016 sec
35	15:28:35	CREATE FUNCTION CalculateAge(dob DATE) RETURNS INT DETERMINISTIC BEGIN RETURN TIMES...	Error Code: 1304 FUNCTION CalculateAge already exists	0.000 sec

Object Info Session

35°

Search

ENG IN 15:28:38 30-06-2026

MySQL Workbench

Local instance MySQL33 - W...

File Edit View Query Database Server Tools Scripting Help

Navigator

Schemas

- Filter objects
- cve_db
- cv2_db
- insure_db
- mydatabase
- nvd
- product
- Tables
- Views
- Stored Procedures

Administration Schemas

Information

No object selected

Query 1 sakila-schema sakila-data SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* SQL File 11* SQL File 12* SQL File 13*

Limit to 1000 rows

```
1
2 SELECT CalculateAge('1990-01-01');
```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Result Grid

CalculateAge('1990-01-01')
36

Result 4 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
35	15:28:35	CREATE FUNCTION CalculateAge(dob DATE) RETURNS INT DETERMINISTIC BEGIN RETURN TIMES...	Error Code: 1304 FUNCTION CalculateAge already exists	0.000 sec
36	15:28:47	SELECT CalculateAge('1990-01-01') LIMIT 0, 1000	1 row(s) returned	0.016 sec / 0.000 sec

Object Info Session

35°

Search

ENG IN 15:28:53 30-06-2026

MySQL Workbench

Local instance MySQL33 - W...

File Edit View Query Database Server Tools Scripting Help

Navigator

Schemas

- Filter objects
- cv2_db
- cv2_db
- insure_db
- mydatabase
- nvd
- product
- Tables
- Views
- Stored Procedures

Administration Schemas

Information

No object selected

Query 1 sakila-schema sakila-data SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* SQL File 11* SQL File 12* SQL File 13*

1 DELIMITER //

2 Execute the selected portion of the script or everything, if there is no selection

3 CREATE FUNCTION HasSufficientBalance(

4 accid INT,

5 amt DECIMAL(10,2)

6)

7 RETURNS BOOLEAN

8 DETERMINISTIC

9 BEGIN

10 DECLARE bal DECIMAL(10,2);

11

12 SELECT Balance

13 INTO bal

14 FROM Accounts

15 WHERE AccountID=accid;

16

17 RETURN bal>amt;

18 END //

19

20 DELIMITER ;

Output

Action Output

#	Time	Action	Message	Duration / Fetch
36	15:28:47	SELECT CalculateAge(1990-01-01) LIMIT 0.1000	1 row(s) returned	0.016 sec / 0.000 sec
37	15:29:10	CREATE FUNCTION HasSufficientBalance(accid INT, amt DECIMAL(10,2)) RETURNS BOOLEAN DE...	0 row(s) affected	0.015 sec

Object Info Session

35°

Search

ENG IN 15:29:12 30-06-2026

MySQL Workbench

Local instance MySQL33 - W...

File Edit View Query Database Server Tools Scripting Help

Navigator

Schemas

- Filter objects
- cv2_db
- cv2_db
- insure_db
- mydatabase
- nvd
- product
- Tables
- Views
- Stored Procedures

Administration Schemas

Information

No object selected

Query 1 sakila-schema sakila-data SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* SQL File 11* SQL File 12* SQL File 13*

1 SELECT HasSufficientBalance(101,500);

2 Execute the selected portion of the script or everything, if there is no selection

Result Grid

HasSufficientBalance(101,500)
1

Result 5 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
37	15:29:10	CREATE FUNCTION HasSufficientBalance(accid INT, amt DECIMAL(10,2)) RETURNS BOOLEAN DE...	0 row(s) affected	0.015 sec
38	15:29:30	SELECT HasSufficientBalance(101,500) LIMIT 0.1000	1 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

35°

Search

ENG IN 15:29:31 30-06-2026

MySQL Workbench

Local instance MySQL33 - W...

File Edit View Query Database Server Tools Scripting Help

Navigator

Filter objects

- cv2_db
- cv2_db
- insure_db
- mydatabase
- nvd
- product
- Tables
- Views
- Stored Procedures

Administration Schemas

Information

No object selected

Query 1 sakila-schema sakila-data SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* SQL File 11* SQL File 12* SQL File 13*

1 DELIMITER //

2 Execute the selected portion of the script or everything, if there is no selection

3 CREATE TRIGGER UpdateCustomerLastModified

4 BEFORE UPDATE

5 ON Customers

6 FOR EACH ROW

7 BEGIN

8 SET NEW.LastModified=NOW();

9 END //

10

11 DELIMITER ;

Output

Action Output

#	Time	Action	Message	Duration / Fetch
38	15:29:30	SELECT HasSufficientBalance(101.500) LIMIT 0.1000	1 row(s) returned	0.000 sec / 0.000 sec
39	15:29:48	CREATE TRIGGER UpdateCustomerLastModified BEFORE UPDATE ON Customers FOR EACH ROW BEGI...	0 row(s) affected	0.015 sec

Object Info Session

35°

Search

ENG IN

15:29:49 30-06-2026

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

MySQL Workbench

Local instance MySQL33 - W...

File Edit View Query Database Server Tools Scripting Help

Navigator

Filter objects

- cv2_db
- cv2_db
- insure_db
- mydatabase
- nvd
- product
- Tables
- Views
- Stored Procedures

Administration Schemas

Information

No object selected

Query 1 sakila-schema sakila-data SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* SQL File 11* SQL File 12* SQL File 13*

1 DELIMITER //

2 Execute the selected portion of the script or everything, if there is no selection

3 CREATE TRIGGER LogTransaction

4 AFTER INSERT

5 ON Transactions

6 FOR EACH ROW

7 BEGIN

8 INSERT INTO AuditLog

9 (TransactionID,ActionDate,ActionTaken)

10 VALUES

11 (NEW.TransactionID,

12 NOW(),

13 'Transaction Inserted');

14 END //

15

16 DELIMITER ;

Output

Action Output

#	Time	Action	Message	Duration / Fetch
39	15:29:48	CREATE TRIGGER UpdateCustomerLastModified BEFORE UPDATE ON Customers FOR EACH ROW BEGI...	0 row(s) affected	0.015 sec
40	15:30:49	CREATE TRIGGER LogTransaction AFTER INSERT ON Transactions FOR EACH ROW BEGIN INSERT L...	0 row(s) affected	0.031 sec

Object Info Session

35°

Search

ENG IN

15:30:53 30-06-2026

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

MySQL Workbench

Local instance MySQL33 - W...

File Edit View Query Database Server Tools Scripting Help

Navigator

Filter objects

SCHEMAS

- ovs_db
- ovs2_db
- insura_db
- mydatabase
- nvd
- product
- Tables
- Views
- Stored Procedures
- Administration
- Schemas

Information

No object selected

Query 1 sakila-schema sakila-data SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* SQL File 11* SQL File 12* SQL File 13* % SQLAdditions

Limit to 1000 rows

```
4  BEGIN
5  -- Execute the selected portion of the script or everything, if there is no selection
6  ON Transactions
7  FOR EACH ROW
8  BEGIN
9
10     DECLARE bal DECIMAL(10,2);
11
12     SELECT Balance
13     INTO bal
14     FROM Accounts
15     WHERE AccountID=NEW.AccountID;
16
17     IF NEW.TransactionType='Withdrawal'
18     AND NEW.Amount>bal THEN
19         SIGNAL SQLSTATE '45000'
20         SET MESSAGE_TEXT='Insufficient Balance';
21     END IF;
22
23     IF NEW.TransactionType='Deposit'
24     AND NEW.Amount<=0 THEN
25         SIGNAL SQLSTATE '45000'
26         SET MESSAGE_TEXT='Invalid Deposit';
27     END IF;
28
29     END //
30
31 DELIMITER ;
```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
40	15:30:49	CREATE TRIGGER LogTransaction AFTER INSERT ON Transactions FOR EACH ROW BEGIN INSERT INTO LogTransaction SET TransactionID=NEW.TransactionID, TransactionType=NEW.TransactionType, Amount=NEW.Amount, AccountID=NEW.AccountID;	0 row(s) affected	0.031 sec
41	15:31:09	CREATE TRIGGER CheckTransactionRules BEFORE INSERT ON Transactions FOR EACH ROW BEGIN IF NEW.TransactionType='Withdrawal' AND NEW.Amount>(SELECT Balance FROM Accounts WHERE AccountID=NEW.AccountID) THEN SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT='Insufficient Balance'; IF NEW.TransactionType='Deposit' AND NEW.Amount<=0 THEN SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT='Invalid Deposit'; END IF; END;	0 row(s) affected	0.047 sec

Object Info Session

35°

Search

ENG IN

15:31:14 30-06-2026